

NSHD - Environmental Exposures - Metadata

The exposure data provided are the result of several years' work from various research teams. Before using any of the exposure estimates in any study we strongly advise that the planned analyses are discussed with the relevant contacts (contact details are provided in the metadata).

Depending on the focus of the research and level of involvement, the authors should be acknowledged in any publication or included as co-authors.

1. Data

File: <i>NSHD_AP_89.csv</i>	(Number of records: 5,362)
<i>NSHD_AP_99.csv</i>	(Number of records: 5,362)
<i>NSHD_AP_06-10.csv</i>	(Number of records: 4,923)

These files contain time relevant air pollution exposure estimates for NSHD participants at their residential address for the 1989, 1999 and 2006-2010 follow-up (i.e. individuals are assigned air pollution concentrations based on their residential address they lived at a specific time). Individual fields and models used to derive the annual average ambient concentrations are described in detail below.

2. Air pollution estimates for the NSHD 1989 follow-up

NSHD_AP_89.csv

Annual average black smoke (BS), sulphur dioxide (SO₂) and nitrogen dioxide (NO₂) concentrations, 1991

Black smoke (BS), sulphur dioxide (SO₂) and nitrogen dioxide (NO₂) maps were modelled based on a land use regression (LUR) model. BS & SO₂ concentrations were modelled as part of the CHES project (Wellcome Trust funded, grant 075883); the NO₂ map is based on input data from the Automatic Urban and Rural Network (AURN), Defra. For details of the models see references Gulliver et al, 2011 (for BS & SO₂) and Gulliver et al, 2013 (for NO₂), citation below.

BS, SO₂ and NO₂ concentrations for the year 1991 were derived from GB-wide air pollution maps. Resolution of BS and SO₂ maps are 100m x 100m, for NO₂ map 200m x 200m. The x,y-coordinates of residential address of NSHD participants in 1989 were overlaid with these maps and the corresponding air pollution concentration of the grid cell were assigned to the coordinate. 1991 is the closest year to the 1989 follow-up for which such information is available.

FIELD	DESCRIPTION
ESCAPE_ID	Unique Identifier
X89	Geocoded X coordinate (British National Grid) based on residential address at time of 1989 follow-up
Y89	Geocoded Y coordinate (British National Grid) based on residential address at time of 1989 follow-up
BS_91	Black smoke; LUR estimate for annual average 1991 ($\mu\text{g}/\text{m}^3$)
SO2_91	Sulphur dioxide; LUR estimate for annual average 1991 ($\mu\text{g}/\text{m}^3$)
NO2_91	Nitrogen dioxide; LUR estimate for annual average 1991 ($\mu\text{g}/\text{m}^3$)

Contact:

BS & SO₂ estimates: Dr Anna Hansell: SAHSU, Imperial College London, UK; email: a.hansell@imperial.ac.uk

NO₂ estimates: Dr John Gulliver: SAHSU, Imperial College London, UK; email: j.gulliver@imperial.ac.uk

References:

Gulliver J, Morris C, Lee K, Vienneau D, Briggs D, Hansell A. 2011. Land use regression modelling to estimate historic (1962-1991) concentrations to black smoke and sulphur dioxide for Great Britain. *Environmental Science and Technology* 45:3526-3532.

Gulliver J, de Hoogh K, Hansell A, Vienneau D. 2013. Development and back-extrapolation of NO₂ land use regression models for historic exposure assessment in Great Britain. *Environmental Science and Technology* 47:7804–7811.

3. Air pollution estimates for the 1999 NSHD follow-up

NSHD_AP_99.csv

3.1 Back extrapolation of ESCAPE LUR model for nitrogen dioxide (NO₂), nitrogen oxides (NO_x) and particulate matter (PM₁₀) concentrations, 1999

These data have been supplied to NSHD previously for use in the LEAP and ESCAPE studies. First, air pollution estimates for the year 2010 were modelled for the address each individual lived at in 1999 using Land Use Regression (LUR) models developed as part of the European Study of Cohorts for Air Pollution Effects (ESCAPE) (<http://www.escapeproject.eu/>), funded under the EU 7th Framework Programme. For details of the models see references Eeftens et al, 2012 and Beelen et al, 2013, citations provided below and modelling of annual average air pollution concentrations, 2010 for the 2006-10 follow-up described in section 4 (page 5).

Based on the modelled 2010 concentrations back-extrapolation to the year 1999 was performed adjusting for time trends based on daily monitoring data from 1996 to 2011. The monitoring site that had the most complete monitoring data across the periods was used for this purpose:

- ESCAPE reference site monitoring period – 16/03/2010 – 01/02/2011
- ESCAPE baseline – 2 years either side of 1 July 1999 (1 July 1998 – 30 June 1999 and 1 July 1999 – 30 June 2000)

Back-extrapolation was only possible for NO₂, NO_x and PM₁₀ and not for all ESCAPE air pollutants as only these three pollutants had representative long-running monitoring sites suitable to use for the back-extrapolation.

Concentrations for 1999 were calculated using both the absolute and the ratio method. The ratio method was used for a published ESCAPE paper on respiratory disease using the NSHD (Cai, 2014) because the spatial contrast in air pollutants was larger in the past and the few available previous studies have used a ratio method (Slama et al, 2007).

Absolute: $C_{\text{extrapolated}} = C_{\text{ESCAPE}} + \text{Diff}_{\text{routine}}$
 Ratio: $C_{\text{extrapolated}} = C_{\text{ESCAPE}} * \text{Ratio}_{\text{routine}}$

Pollutant	C _{Routine ESCAPE}	C _{Routine Baseline}	Diff _{Routine}	Ratio _{Routine}
NO ₂	11.1	14.0	2.9	14/11.1 = 1.27
NO _x	13.1	18.6	5.5	18.6/13.1 = 1.42
PM ₁₀	14.9	18.2	3.3	18.2/14.9 = 1.22

For full details please see:

http://www.escapeproject.eu/manuals/Procedure_for_extrapolation_back_in_time.pdf

Please note that the ESCAPE estimates for PM₁₀ are valid up to 400km from the monitoring area, which was in Greater London, but it is unclear how good the estimates are outside this area. All addresses which are more than 400km away from Greater London are not assigned a PM₁₀ estimate concentrations and have missing data (.) instead.

Please also note that if your study requires use of air pollution estimates for the 2006-10 follow-up and additionally the 1999 follow-up we recommend not using ESCAPE for both time-points, i.e. ESCAPE (for 2006-10, see section 4) together with back-extrapolated ESCAPE (for 1999, as here, section 3.1) as these estimates are not independent of each other – back-extrapolated ESCAPE estimates were derived from the original ESCAPE estimates. An alternative to back-extrapolated ESCAPE estimates in this situation is the RGI estimates for 2001 (detailed below, section 3.2).

FIELD	DESCRIPTION
ESCAPE_ID	Unique Identifier
X99	Geocoded X coordinate (British National Grid) based on residential address at time of 1999 follow-up
Y99	Geocoded Y coordinate (British National Grid) based on residential address

	at time of 1999 follow-up
no2_99_a	Nitrogen dioxide; back extrapolation to 1999 based on LUR estimate for 2010 ($\mu\text{g}/\text{m}^3$) using the absolute method
no2_99_r	Nitrogen dioxide; back extrapolation to 1999 based on LUR estimate for 2010 ($\mu\text{g}/\text{m}^3$) using the ratio method
nox_99_a	Nitrogen oxides; ; back extrapolation to 1999 based on LUR estimate for 2010 ($\mu\text{g}/\text{m}^3$) using the absolute method
nox_99_r	Nitrogen oxides; back extrapolation to 1999 based on LUR estimate for 2010 ($\mu\text{g}/\text{m}^3$) using the ratio method
pm10_99_a	PM ₁₀ (particulate matter with diameter $\leq 10\mu\text{m}$); back extrapolation to 1999 based on LUR estimate for 2010 ($\mu\text{g}/\text{m}^3$) using the absolute method
pm10_99_r	PM ₁₀ (particulate matter with diameter $\leq 10\mu\text{m}$); back extrapolation to 1999 based on LUR estimate for 2010 ($\mu\text{g}/\text{m}^3$) using the ratio method

Contact:

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Dr Kees de Hoogh, Swiss Tropical and Public Health Institute, CH (who led the ESCAPE exposure assessment for the UK); email: c.dehoogh@unibas.ch

References:

Beelen R, Hoek G, Vienneau D, Eeftens M, Dimakopoulou K, Pedeli X, Tsai M-Y, Künzli N, Schikowski T, Marcon A, Eriksen KT, Raaschou-Nielsen O, Stephanou E, Patelarou E, Lanki T, Yli-Tuomi T, Declercq C, Falq G, Stempfelet M, Birk M, Cyrus J, von Klot S, Nádor S, Varró MJ, Dédélé A, Gražulevičienė R, Mölter A, Lindley S, Madsen C, Cesaroni G, Ranzi A, Badaloni C, Hoffmann B, Nonnemacher M, Krämer U, Kuhlbusch T, Cirach M, de Nazelle A, Nieuwenhuijsen M, Bellander T, Korek M, Olsson D, Strömgren M, Dons E, Jerrett M, Fischer P, Wang M, Brunekreef B, de Hoogh K. 2013. Development of NO₂ and NO_x land use regression models for estimating air pollution exposure in 36 study areas in Europe – The ESCAPE project. *Atmospheric Environment* 72: 10-23.

Cai Y, Schikowski T, Adam M, Buschka A, Carsin AE, Jacquemin B, Marcon A, Sanchez M, Vierkötter A, Al-Kanaani Z, Beelen R, Birk M, Brunekreef B, Cirach M, Clavel-Chapelon F, Declercq C, de Hoogh K, de Nazelle A, Ducret-Stich RE, Valeria Ferretti V, Forsberg B, Gerbase MW, Hardy R, Heinrich J, Hoek G, Jarvis D, Keidel D, Kuh D, Nieuwenhuijsen MJ, Ragettli MS, Ranzi A, Rochat T, Schindler C, Sugiri D, Temam S, Tsai MY, Varraso R, Kauffmann F, Krämer U, Sunyer J, Künzli N, Probst-Hensch N, Hansell AL. 2014. Cross-sectional associations between air pollution and chronic bronchitis: an ESCAPE meta-analysis across five cohorts. *Thorax* 69(11):1005-1014.

Eeftens M, Beelen R, de Hoogh K, Bellander T, Cesaroni G, Cirach M, Declercq C, Dédélé A, Dons E, de Nazelle A, Dimakopoulou K, Eriksen K, Falq G, Fischer P, Galassi C, Gražulevičienė, Heinrich J, Hoffmann B, Jerrett M, Keidel D, Korek M, Lanki T, Lindley S, Madsen C, Mölter A, Nádor G, Nieuwenhuijsen M, Nonnemacher M, Pedeli X, Raaschou-Nielsen O, Patelarou E, Quass U, Ranzi A, Schindler C, Stempfelet M, Stephanou E, Sugiri D, Tsai M-Y, Yli-Tuomi T, Varró MJ, Vienneau D, von Klot S, Wolf K, Brunekreef B, Hoek G. 2012. Development of Land Use Regression models for PM_{2.5}, PM_{2.5} absorbance, PM₁₀ and PM_{COARSE} in 20 European study areas; results of the ESCAPE project. *Environmental Science & Technology* 46(20): 11195–11205.

Slama R, Morgenstern V, Cyrus J, Zutavern A, Herbarth O, Wichmann H-E, Heinrich J, LISA Study Group. 2007. Traffic-related atmospheric pollutants levels during pregnancy and offspring's term birth weight: A study relying on a land-use regression exposure model. *Environmental Health Perspectives* 115: 1283-1292.

ESCAPE air pollution modelling manual for back-extrapolation

http://www.escapeproject.eu/manuals/Procedure_for_extrapolation_back_in_time.pdf

3.2 Nitrogen dioxide (NO₂) and particulate matter (PM₁₀) concentrations, 2001

NO₂ and PM₁₀ maps were modelled based on a land use regression (LUR) model as part of the RGI project (grant RGI-137, Dutch program Ruimte voor Geoinformatie). For details of the model see reference Vienneau et al., 2009, citation below.

NO₂ and PM₁₀ concentrations for the year 2001 were derived from GB-wide air pollution maps (resolution 100m x 100m). X,y-coordinates of NSHD participants were overlaid with these maps and the corresponding air pollution concentration of the 100m x 100m grid cell were assigned to the coordinate. 2001 is the closest year to the 1999 follow-up for which such information is available.

FIELD	DESCRIPTION
NO2_01	Nitrogen dioxide; LUR estimate for annual average 2001 (µg/m ³)
PM10_01	PM ₁₀ (particulate matter with diameter ≤10m); LUR estimate for annual average 2001 (µg/m ³)

Contact:

Dr Danielle Vienneau: Swiss Tropical and Public Health Institute, CH; email: Danielle.Vienneau@unibas.ch

References:

Vienneau D, de Hoogh K, Beelen R, Fischer P, Hoek G, Briggs D. 2010. Comparison of land-use regression models between Great Britain and the Netherlands. *Atmospheric Environment* 44: 688-696.

4. Air pollution estimates for the 2006-2010 NSHD follow-up

NSHD_AP_06-10.csv

Annual average air pollution concentrations, 2010 (NO₂, NO_x, PM_{2.5}, PM_{2.5} absorbance, PM₁₀ and PM_{COARSE})

Annual average concentrations for the year 2010 were modelled for each individual's address during the 2006-10 follow-up using a Land Use Regression (LUR) model developed as part of the European Study of Cohorts for Air Pollution Effects (ESCAPE) (<http://www.escapeproject.eu/>), funded under the EU 7th Framework Programme. For details of the models see references Eeftens et al, 2012 and Beelen et al 2013, citations provided below.

Land Use Regression (LUR) was used to estimate ambient air pollution concentrations at the home addresses of the NSHD participants. LUR combines the actual measurement of air pollution done between 26 Jan 2010 - 18 January 2011 and the development of stochastic models using predictor variables such as traffic variables, population density and meteorology derived with Geographical Information Systems (GIS) to develop statistical models using linear regression. The model is then applied to a large number of unsampled locations in the study area. This technique is widely used in epidemiological research to investigate possible links between air pollution and health effects (for a review of LUR models to assess air pollution, see Hoek et al., 2008).

Air pollution estimates are representative for the year 2010. Traffic variables were calculated during the course of LUR.

Please note that the ESCAPE estimates for PM₁₀, PM_{2.5}, PM_{2.5} absorbance and PM coarse are valid up to 400km from the monitoring area, which was in Greater London, but it is unclear how good the estimates are outside this area. All addresses which are more than 400km away from Greater London are not assigned a PM₁₀ estimate concentrations and have missing data (.) instead.

Please note that some NSHD participant changed address around the 2006-2010 follow-up period which results in multiple entries of individuals (ESCAPE_ID) because air pollution 2010 annual average air pollution concentrations were modelled for all addresses a NSHD participant lived at between 2004 and 2011.

FIELD	DESCRIPTION
ESCAPE_ID	Unique Identifier
X06_10	Geocoded X coordinate (British National Grid) based on residential address at time of 2006-10 follow-up
Y06_10	Geocoded Y coordinate (British National Grid) based on residential address at time of 2006-10 follow-up
no2_10	Nitrogen dioxide; LUR estimate for annual average 2010 ($\mu\text{g}/\text{m}^3$)
nox_10	Nitrogen oxides; LUR estimate for annual average 2010 ($\mu\text{g}/\text{m}^3$)
pm10_10	PM ₁₀ (particulate matter with diameter $\leq 10\mu\text{m}$); LUR estimate for annual average 2010 ($\mu\text{g}/\text{m}^3$)
pm25_10	PM _{2.5} (particulate matter with diameter $\leq 2.5\mu\text{m}$); LUR estimate for annual average 2010 ($\mu\text{g}/\text{m}^3$)
pm25abs_10	PM _{2.5} absorbance (measurement of the blackness of PM _{2.5} filters; a proxy for elemental carbon, which is the dominant light absorbing substance); LUR estimate for annual average 2010 (m-1)
pmcoarse_10	PM coarse (particulate matter 2.5-10 μm); LUR estimate for annual average 2010 ($\mu\text{g}/\text{m}^3$)
Trafnear	Traffic intensity on the nearest road based upon local road network
Distinvnear1	Inverse distance to the nearest road based upon local road network

Trafmajor	Traffic intensity on the nearest major road defined as a road with traffic intensity > 5,000 vehicles / day based upon a local road network
Distinvmajor1	Inverse distance to the nearest major road defined as a road with traffic intensity > 5,000 vehicles / day based upon a local road network
Trafmajorload100	Total traffic load (intensity*length) on major roads in a 100m buffer based upon local road network
Majorroad	Indicator variable indicating whether a coordinate is within 50m of a class 1 or 2 type road and/or within 100m of a class 0 road (=motorway), based upon central road network
Majorroadlength100	Sum of road length of major roads defined as class 0, 1 or 2 (and possibly classes 3 or 4 based upon local knowledge) from the central road network within a 100m buffer

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Hoek G, Beelen R, de Hoogh K, Vienneau D, Fischer P, Briggs D. 2008. A review of land use regression models to assess spatial variation of outdoor air pollution. *Atmospheric Environment* 42: 7561-7578.